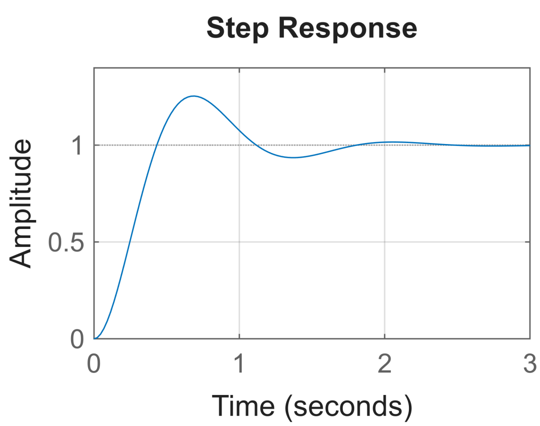
```
%L1.a. Time Domain Analysis

%Problem 1:
%Obtain the unit step response of the system whose
%closed loop transfer function is given by (C(s))/(R(s)) = 25/(s ^ 2 + 4s + 25)

num=[25]
```

```
num =
25
```

```
den=[1 4 25]
```

```
den = 1×3
     1     4    25
```

```
figure(1);
step(num,den)
grid on
```




```
%L1.a. Time Domain Analysis

%Problem 2
%500(s+2)(s+5)(s+6)/s(s+8)(s+10)(s+12)

s=tf('s');
G=(500*(s+2)*(s+5)*(s+6))/(s*(s+8)*(s+10)*(s+12))
```

```
G =

  500 s^3 + 6500 s^2 + 26000 s + 30000
  ------------------------------------
     s^4 + 30 s^3 + 296 s^2 + 960 s

Continuous-time transfer function.
Model Properties
```

```
%stability
T=feedback(G,1)
```

```
T =

     500 s^3 + 6500 s^2 + 26000 s + 30000
  ------------------------------------------
  s^4 + 530 s^3 + 6796 s^2 + 26960 s + 30000

Continuous-time transfer function.
Model Properties
```

```
poles=pole(T)
```

```
poles = 4×1
 -516.9544
   -5.7623
   -5.4278
   -1.8554
```

```
%steady state error
%step input
Kp=dcgain(G)
```

```
Kp =
Inf
```

```
essKp=1/(1+Kp)
```

```
essKp =
0
```

```
%ramp input
Kv=dcgain(minreal(s*G))
```

```
Kv =
31.2500
```

```
essKv=1/Kv
```

```
essKv =
0.0320
```

```
%parabolic input
Ka=dcgain(minreal((s^2)*G))
```

```
Ka =
0
```

```
essKa=1/Ka
```

```
essKa =
Inf
```

```
%L1.a. Time Domain Analysis

%Problem 3:
%Gain design to meet a steady state error specification
%(K(s + 5))/(s(s + 6)(s + 7)(s + 8))
```

```
%ess=0.1: desired steady state velocity error

s=tf('s');
G=(s+5)/(s*(s+6)*(s+7)*(s+8))
```

```
G =

              s + 5
  ------------------------------
  s^4 + 21 s^3 + 146 s^2 + 336 s

Continuous-time transfer function.
Model Properties
```

```
ess=0.1
```

```
ess =
0.1000
```

```
Kv=dcgain(minreal(s*G))
```

```
Kv =
0.0149
```

```
K=1/(ess*Kv)
```

```
K =
672.0000
```

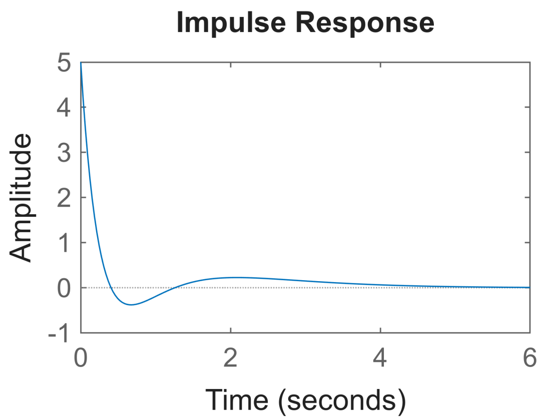
```
Gnew=feedback(K*G,1)
```

```
Gnew =

               672 s + 3360
  -------------------------------------
  s^4 + 21 s^3 + 146 s^2 + 1008 s + 3360

Continuous-time transfer function.
Model Properties
```

```
pole(Gnew)
```

```
ans = 4×1 complex
 -14.6663 + 0.0000i
  -0.6909 + 6.7667i
  -0.6909 - 6.7667i
  -4.9518 + 0.0000i
```

```
%L1.a. Time Domain Analysis

% Exercise 1:
% Obtain the impulse response of the closed loop transfer function
% (C(s))/(R(s)) = (5s ^ 2 + 3s + 6)/(s ^ 3 + 6s ^ 2 + 11s + 6)

n=[5 3 6];
d=[1 6 11 6];
impulse(n,d)
```




```
%L1.a. Time Domain Analysis

% Exercise 2:
% Obtain the unit step responses of the following systems whose closed
% loop transfer function. Plot their step responses curves in one figure window.
and
% (C(s))/(R(s)) = 1/(s ^ 2 + 0.5s + 1)
% (C(s))/(R(s)) = 1/(s ^ 2 + 0.5s + 4)

clc
clear
close all

figure(1)

t=0:0.1:20;
n1=[0 0 1];
d1=[1 0.5 1];
n2=[0 0 1];
d2=[1 0.5 4];

[y1,x1,t]=step(n1,d1,t)
```

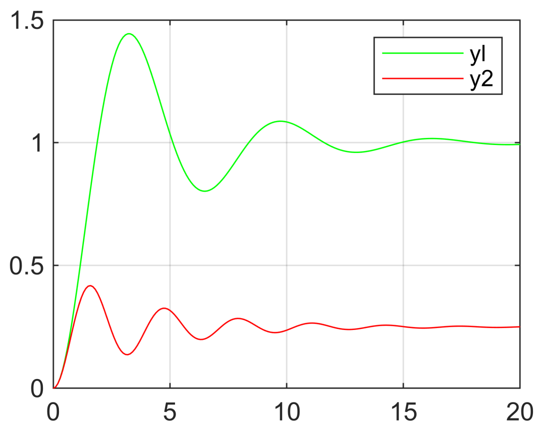
```
y1 = 201×1
         0
    0.0049
    0.0193
    0.0425
    0.0739
    0.1129
    0.1585
    0.2102
    0.2670
    0.3282
      ⋮

x1 =

     []
```

```
t = 1×201
        0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ⋯
```

```
[y2,x2,t]=step(n2,d2,t)
```

```
y2 = 201×1
        0
   0.0049
   0.0191
   0.0416
   0.0711
   0.1061
   0.1450
   0.1861
   0.2276
   0.2680
     ⋮
x2 =

    []
t = 1×201
        0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ⋯
```

```
plot(t,y1,'g',t,y2,'r')
legend('yl', 'y2')
grid on
```




```
%L1.a. Time Domain Analysis

% Exercise 3
% For the following closed loop transfer functions obtain the step/impulse/ramp
responses of the given system. ii) And also obtain the time domain specification
from step response and verify result theoretically.
% (C(s))/(R(s)) = (wn^2)/(s ^ 2 + 2*zeta*wn*s + wn^2)
% where zeta = 0.6 and wn = 8

wn=8;
zeta=0.6;
```

```
num=[0 0 wn^2]
```

```
num = 1×3
     0     0    64
```

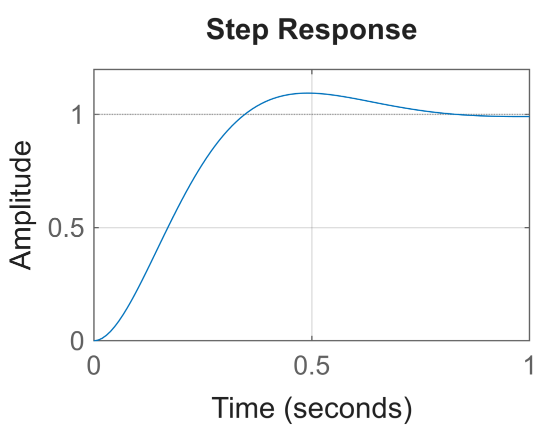
```
den=[1 2*zeta*wn wn^2]
```

```
den = 1×3
    1.0000    9.6000   64.0000
```

```
sys=tf(num,den)
```

```
sys =
 
         64
  ----------------
  s^2 + 9.6 s + 64
 
Continuous-time transfer function.
Model Properties
```

```
figure(1)
step(num,den)
grid on
```

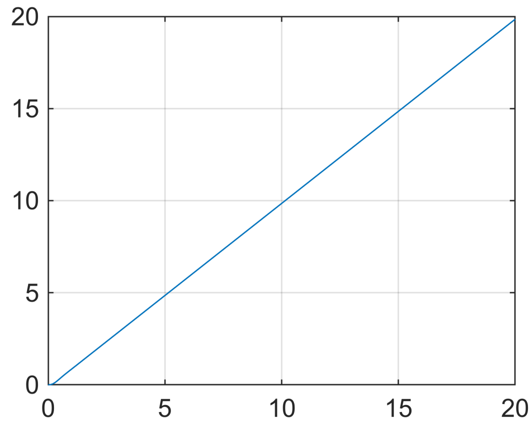



```
t=0:0.1:20;
ramp=t;
[yramp,tramp]=lsim(sys,t,t)
```

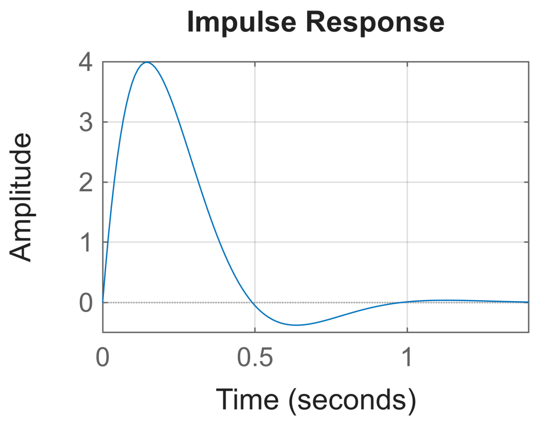
```
yramp = 201×1
         0
    0.0083
    0.0504
    0.1281
    0.2281
    0.3366
    0.4451
    0.5503
    0.6521
    0.7520
      ⋮

tramp = 201×1
         0
    0.1000
```

```
    0.2000
    0.3000
    0.4000
    0.5000
    0.6000
    0.7000
    0.8000
    0.9000
      ⋮
```

```
figure(2)
plot(tramp,yramp)
grid on
```




```
figure(3)
impulse(num,den)
grid on
```




```
%L1.a. Time Domain Analysis

% Exercise 4:
% Determine the step, ramp, and parabolic error constants and the steady state
error
```

```
% for the unity negative feedback control system with following open loop transfer
function. Also plot the respective time domain responses.
% 5s² +2s+1 G(s)/s(s+2)(s +8)(s +45)

s = tf('s');
G = (5*s^2 + 2*s + 1)/(s*(s+2)*(s+8)*(s+45));

kp=dcgain(G)
```

```
kp =
Inf
```

```
esskp=1/(1+kp)
```

```
esskp =
0
```

```
kv=dcgain(minreal(s*G))
```

```
kv =
0.0014
```

```
esskv=1/kv
```

```
esskv =
720.0000
```

```
ka=dcgain(minreal((s^2)*G))
```

```
ka =
0
```

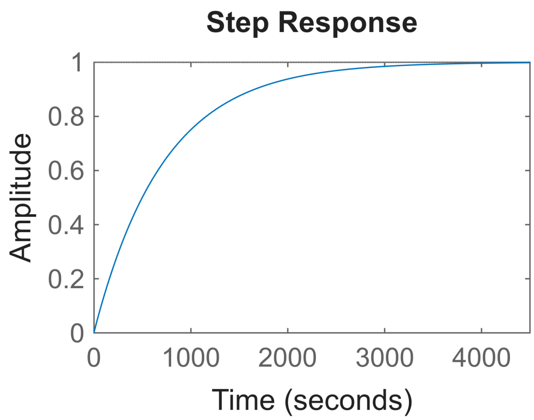
```
esska=1/ka
```

```
esska =
Inf
```

```
T=feedback(G,1)
```

```
T =

           5 s^2 + 2 s + 1
  ----------------------------------
  s^4 + 55 s^3 + 471 s^2 + 722 s + 1

Continuous-time transfer function.
Model Properties
```

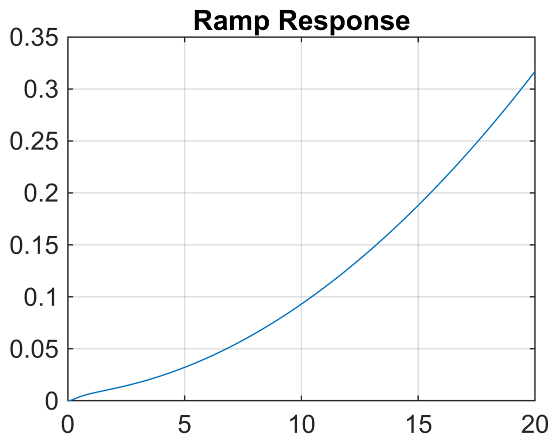
```
figure(1)
step(T)
```




```
grid on

t=0:0.1:20;
ramp=t;
figure(2)
[yr,tr]=lsim(T,t,t)
```

```
yr = 201×1
         0
    0.0003
    0.0010
    0.0020
    0.0029
    0.0037
    0.0045
    0.0052
    0.0059
    0.0065
      ⋮
tr = 201×1
         0
    0.1000
    0.2000
    0.3000
    0.4000
    0.5000
    0.6000
    0.7000
    0.8000
    0.9000
      ⋮
```

```
plot(tr,yr)
title("Ramp Response")
grid on
```



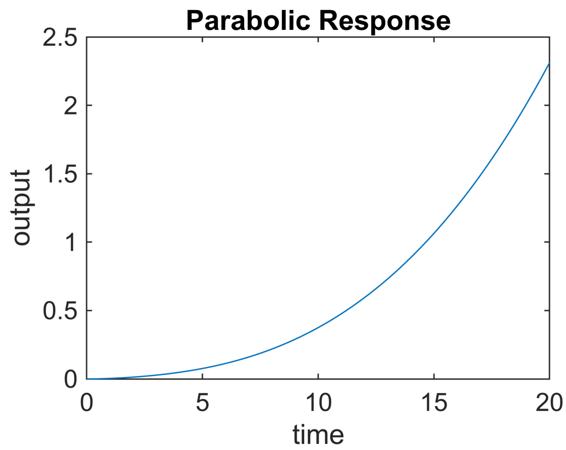

```
figure(3)
para=(t.^2)/2
```

```
para = 1×201
         0    0.0050    0.0200    0.0450    0.0800    0.1250    0.1800    0.2450 ···
```

```
[pr,tr]=lsim(T,para,t)
```

```
pr = 201×1
         0
    0.0000
    0.0001
    0.0002
    0.0005
    0.0008
    0.0012
    0.0017
    0.0023
    0.0029
      ⋮

tr = 201×1
         0
    0.1000
    0.2000
    0.3000
    0.4000
    0.5000
    0.6000
    0.7000
    0.8000
    0.9000
      ⋮
```

```
plot(tr,pr)
title("Parabolic Response")
xlabel("time")
ylabel("output")
```




```
impulse(T)
grid on
```